

Tutorial - 02

A moving Streaming Service intends to classify movies into categories.

Action & Romance using a KNN Classifier.

Input features:

Action Scenes, Count - no. of action sequences

Romance Scenes Count - no. of romantic

Target Variables

Genre (Action = A, Romance = R)

<u>Mid</u>	<u>Action Scene</u>	<u>Romance</u>	<u>Genre</u>
1	15	2	A
2	1	12	R
3	18	1	A
4	2	14	R
5	14	3	A
6	3	11	R
7	16	2	R
8	2	13	R

Test Dataset [with actual labels]:

<u>Mid</u>	<u>Action Score</u>	<u>Romance</u>	<u>Genre</u>
T ₁	10	5	A
T ₂	4	10	R
T ₃	17	2	A

Euclidean Formula:

$$D = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

$$(1, 5) \rightarrow \sqrt{(10-1)^2 + (5-2)^2} \Rightarrow \sqrt{25+9} = \sqrt{34} = 5.83$$

$$(1, 12) \rightarrow \sqrt{(10-1)^2 + (5-12)^2} \Rightarrow \sqrt{81+49} = \sqrt{130} = 11.4$$

$$(18, 1) \rightarrow \sqrt{(10-18)^2 + (5-1)^2} \Rightarrow \sqrt{64+16} = \sqrt{80} = 8.94$$

$$(2, 14) \rightarrow \sqrt{(10-2)^2 + (5-14)^2} \Rightarrow \sqrt{64+81} = \sqrt{145} = 12.04$$

$$(14, 3) \rightarrow \sqrt{(10-14)^2 + (5-3)^2} = \sqrt{16+4} = \sqrt{20} = 4.47$$

$$(3, 11) \rightarrow \sqrt{(10-3)^2 + (5-11)^2} = \sqrt{49+36} = \sqrt{85} = 9.21$$

$$(6, 2) \rightarrow \sqrt{(10-6)^2 + (5-2)^2} = \sqrt{36+9} = \sqrt{45} = 6.70$$

$$(2, 13) \rightarrow \sqrt{(10-2)^2 + (5-13)^2} \Rightarrow \sqrt{64+64} = \sqrt{128} \Rightarrow 11.31$$

Movied	Distance	Genre
5	4.47	A
1	5.38	A
7	6.70	A

Classification of T_1

When $K=1$ Nearest Neighbours (5, 1, 7) Gene A

Prediction (A)

(x, y) $(4, 10)$

$$(15, 2) = \sqrt{(4-15)^2 + (10-2)^2} = \sqrt{121 + 64} = \sqrt{185} = 13.6$$

$$(1, 12) = \sqrt{(4-1)^2 + (10-12)^2} = \sqrt{9 + 4} = \sqrt{13} = 3.60$$

$$(18, 1) = \sqrt{(4-18)^2 + (10-1)^2} = \sqrt{196 + 81} = \sqrt{277} = 16.6$$

$$(2, 14) = \sqrt{(4-2)^2 + (10-14)^2} = \sqrt{4 + 16} = \sqrt{20} = 4.47$$

$$(14, 3) = \sqrt{(4-14)^2 + (10-3)^2} = \sqrt{100 + 49} = \sqrt{149} = 12.20$$

$$(3, 11) = \sqrt{(4-3)^2 + (10-11)^2} = \sqrt{1 + 1} = \sqrt{2} = 1.41$$

$$(16, 2) = \sqrt{(4-16)^2 + (10-2)^2} = \sqrt{144 + 64} = \sqrt{208} = 14.4$$

$$(2, 18) = \sqrt{(4-2)^2 + (10-18)^2} = \sqrt{4 + 64} = \sqrt{68} = 8.24$$

Table 12.

	Dist.	Gene
6	1.41	R
2	3.60	R
8	3.60	R

(7,4)(17,2)

$$(15,2) = \sqrt{(17-15)^2 + (2-2)^2} = \sqrt{4+0} = 2$$

$$(1,12) = \sqrt{(17-1)^2 + (2-12)^2} = \sqrt{236+100} = 18.8$$

$$(18,1) = \sqrt{(17-18)^2 + (2-1)^2} = \sqrt{1+1} = \sqrt{2} = 1.4$$

$$(2,14) = \sqrt{(17-2)^2 + (2-14)^2} = \sqrt{225+144} = \sqrt{369} = 19.2$$

$$(14,3) = \sqrt{(17-14)^2 + (2-3)^2} = \sqrt{9+1} = \sqrt{10} = 3.16$$

$$(3,11) = \sqrt{(17-3)^2 + (2-11)^2} = \sqrt{196+81} = \sqrt{277} = 16.6$$

$$(16,2) = \sqrt{(17-16)^2 + (2-2)^2} = \sqrt{1+0} = \sqrt{1} = 1$$

$$(2,13) = \sqrt{(17-2)^2 + (2-13)^2} = \sqrt{225+121} = \sqrt{346} = 18.60$$

Movie ID	Distance	Genre
7	1	A
3	1.4	A
1	2	A

Actual labels Vs predicted labels

TID	Actual	Predicted (k=1)	(k=2)	(k=3)
T ₁	A	A	A	A
T ₂	R	R	R	R
T ₃	A	A	A	A

- All predictions match Actual labels.

3. Accuracy = $\frac{\text{Correct predictions}}{\text{Total predictions}}$

$$= \frac{3}{3} = 1 \times 100 = 100\% \rightarrow \text{Accuracy}$$

4. Confusion Matrix:

Action $\rightarrow$ Positive

Romance $\rightarrow$ Negative

	A	R
A	2	0
R	0	1

$$TP = 2$$

$$FN = 0$$

$$FP = 0$$

$$TN = 1$$

5. Precision = $\frac{TP}{(TP+FP)} = \frac{2}{2} = 1$

$$\text{Precision} = 100\%$$

F1 Score:

$$F_1 = \frac{2 \times (\text{precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})}$$

$$F_1 = \frac{2 \times (1 \times 1)}{1 + 1} = 1$$

$$F_1 \text{ Score} = 100\%$$